Quantification: 543_z13
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	11
interior lacuna count	11
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 11)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1862	1736	823.012	px^2
major_axis_length_px	lacuna shape	78.3615	74.0218	22.3719	px
minor_axis_length_px	lacuna shape	30.8137	31.6227	6.9101	px
aspect_ratio	lacuna shape	2.5873	2.6086	0.6752	unitless
eccentricity	lacuna shape	0.9006	0.9236	0.0676	unitless
solidity	lacuna shape	0.9037	0.9231	0.0388	unitless
roots_count	headline	13.0909	12	3.7538	count
ring_length_r30_px	headline	419.182	418	69.4951	px
ring_length_r60_px	reported	1070.27	1031	141.355	px
owned_length_px	ownership-dependent	3633.46	3691.75	1254.1	px
edge_count	ownership-dependent	167.273	144	56.0537	count
mean_edge_length_px	ownership-dependent	21.7346	21.2921	1.9708	px
perimeter_px	normalised	207.479	205.037	54.0901	px
ring_area_r30_px2	normalised	8447.82	8219	1466.9	px^2
ring_area_r60_px2	normalised	21999.7	21794	3275.21	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	1	1	0	unitless
ring_density_r30	normalised	0.0501265	0.0496411	0.00734064	px^-1
ring_density_r60	normalised	0.0490432	0.0492474	0.00554488	px^-1
roots_per_100px_perimeter	normalised	6.4973	5.8526	2.0099	per 100 px
ring_attached_length_r30_px	network v2	391	374	75.4877	px
ring_attached_length_r60_px	network v2	942.091	898	189.369	px
ring_length_w_r30_px	network v2	476.329	467.061	78.2104	px
ring_length_w_r60_px	network v2	1215.16	1149.67	162.175	px
sholl_crossings_r10	network v2	10.0909	10	2.3433	count
sholl_crossings_r20	network v2	13.7273	14	2.9695	count
sholl_crossings_r30	network v2	14.1818	14	2.7136	count
loop_count	loops, report only	18.0909	16	9.9444	count
loop_edge_share	loops, report only	0.3955	0.4314	0.1049	share
ring_width_mean_r30_px	width	7.2074	7.3203	0.4981	px

Per field
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measure	value	units
analysed_area_px2	1.02809e+06	px^2
total_skeleton_length_px	44193	px
canalicular_length_density_per_px	0.0429854	px^-1
junction_count	1095	count
junction_density_per_px2	0.00106508	px^-2
lacuna_count	11	count

lacunae_per_px2	1.07e-05	px^-2
skeleton_component_count	191	count
mean_component_length_px	231.377	px
median_component_length_px	22	px
field_length_density_w_per_px	0.0488961	px^-1
field_density_without_flagged_per_px	0.0429854	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	256	px
unattached_component_count	13	count
canalicular_length_density_attached_per_px	0.0429854	px^-1
field_length_density_with_unattached_per_px	0.0432344	px^-1
loop_count	248	count
loop_edge_share	0.4336	share
edge_count	2235	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.8083	px
width_median_px	8	px
width_p10_px	4.4721	px
width_p90_px	10.198	px
width_pixels_used	44193	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2019	79.1535	33.2822	2.3783	0.9073	0.9417	0.4378	97.9678	270.286	False
2	839	61.6665	17.863	3.4522	0.9571	0.906	0.068	131.349	828.439	False
3	1736	73.3356	33.3815	2.1969	0.8904	0.8374	0.2312	218.82	919.248	False
4	1138	46.8317	31.6227	1.481	0.7376	0.9252	0.396	263.104	340.498	False
5	2267	87.9083	33.2852	2.6411	0.9255	0.9473	0.0042	459.621	481.093	False
6	860	45.4745	26.4762	1.7176	0.813	0.854	0.07	475.221	748.086	False
7	2150	103.918	27.5311	3.7745	0.9643	0.9231	0.2711	546.673	873.734	False
8	3069	109.359	37.084	2.9489	0.9407	0.9099	0.049	642.67	188.33	False
9	1616	74.0218	28.3763	2.6086	0.9236	0.9245	-0.0477	731.536	630.238	False
10	1438	72.5092	25.5367	2.8394	0.9359	0.9236	0.0953	793.137	478.379	False
11	3350	107.799	44.5116	2.4218	0.9108	0.8475	-0.088	854.917	288.104	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	12	357	975	4395.66	215	20.4449	7.4298
2	17	422	1031	5635.99	242	23.2892	6.9389
3	16	408	970	2352.81	114	20.6387	6.864
4	7	305	953	2533.76	119	21.2921	7.6428
5	12	418	1155	5107.46	233	21.9204	7.6884
6	8	415	1014	2205.93	121	18.2308	7.3203
7	20	525	1259	4687.42	222	21.1145	6.4672
8	14	519	1198	3691.75	144	25.6371	7.1878

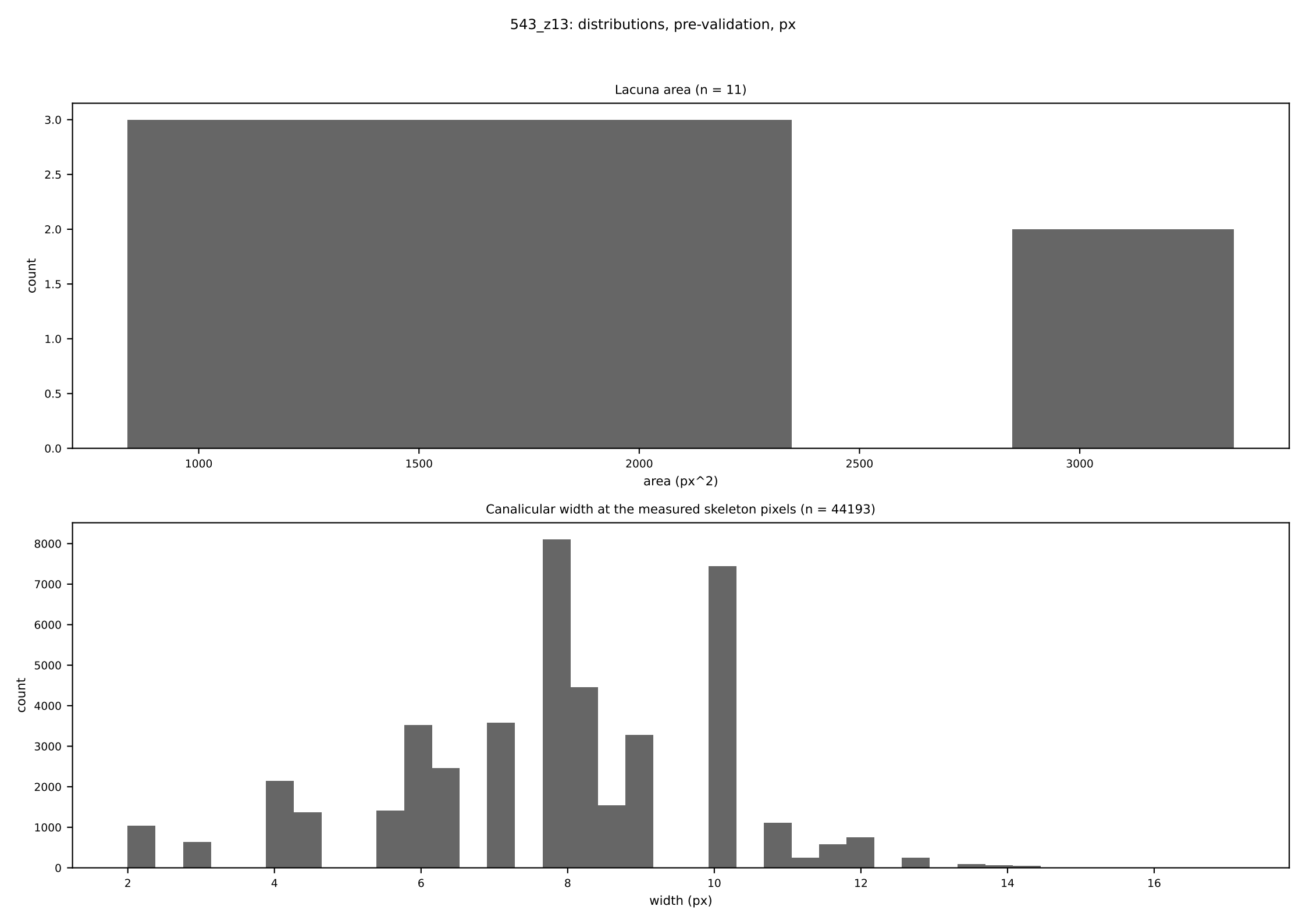
9	12	337	848	2081.56	89	23.3883	7.3832
10	14	429	1068	3111.51	137	22.7118	6.3959
11	12	476	1302	4164.16	204	20.4125	7.9628

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	205.037	8499	22530	1	1	0.0420049	0.0432756	5.8526
2	151.297	6982	17707	1	1	0.0604411	0.0582256	11.2362
3	205.037	8219	20045	1	1	0.0496411	0.0483911	7.8035
4	139.681	6734	18999	1	1	0.0452926	0.0501605	5.0114
5	223.823	8900	23453	1	1	0.0469663	0.0492474	5.3614
6	141.338	6516	17817	1	1	0.0636894	0.0569119	5.6602
7	254.936	9810	24446	1	1	0.0535168	0.0515013	7.8451
8	278.593	10387	26403	1	1	0.0499663	0.0453736	5.0253
9	195.723	8083	21794	1	1	0.0416924	0.0389098	6.1311
10	184.995	7835	21305	1	1	0.0547543	0.0501291	7.5678
11	301.806	10961	27498	1	1	0.0434267	0.0473489	3.9761

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	355	909	422.274	1129.74	11	12	10
2	397	898	466.664	1149.67	9	14	15
3	334	640	467.061	1103.93	12	14	13
4	300	858	351.078	1078.61	6	9	13
5	374	1022	467.706	1290.65	10	14	17
6	370	824	464.392	1138.12	7	13	14
7	525	1182	599.487	1432.66	13	18	15
8	505	1173	593.416	1377.38	13	17	18
9	297	774	381.907	968.808	9	9	11
10	393	844	483.605	1209.21	9	14	12
11	451	1239	542.032	1487.97	12	17	18


543_z13: distributions, pre-validation, px
Lacuna area (n = 11)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 44193)
count
width (px)